

Homework 8

Due: November 24, 2025

Problem 1. A Weibull distribution with parameters $\alpha > 0$ and $\beta > 0$ has pdf given by

$$f(t) = \alpha\beta^{-\alpha}t^{\alpha-1}e^{-(t/\beta)^\alpha}, t > 0.$$

Like the exponential distribution, the Weibull distributions are sometimes used to model the time till an event takes place (note that when $\alpha = 1$, the pdf reduces to an exponential pdf with parameter β^{-1}).

You gathered the following data on the Google search engine response times (in seconds):

0.2, 0.01, 0.02, 0.06, 0.1, 0.05, 0.24, 0.17, 0.03, 0.12, 0.06, 0.08, 0.13, 0.22, 0.14.

Suppose that the response times of Google search engine are independent, and have a Weibull distribution with $\alpha = 2$. Find the maximum likelihood estimator of β .

Problem 2. River floods are often measured by their discharges (in units of feet cubed per second). The value v is said to be the value of a 100-year flood if

$$\mathbb{P}(D \geq v) = 0.01,$$

where D is the discharge of the largest flood in a randomly chosen year. The following table gives the flood discharges of the largest floods of the Blackstone River in Woonsocket, Rhode Island, in each of the years from 1929 to 1965. Assuming that these discharges follow a lognormal distribution (i.e., $\log D$ follows a normal distribution), estimate the value of a 100-year flood.

Annual Floods of the Blackstone River (1929–1965)

Year	Flood Discharge (ft ³ /s)
1929	4,570
1930	1,970
1931	8,220
1932	4,590
1933	5,780
1934	6,560
1935	7,500
1936	15,000
1937	6,340
1938	15,100
1939	3,840
1940	5,860
1941	4,480
1942	5,350
1943	5,310
1944	3,830
1945	3,410
1946	3,830
1947	3,150
1948	5,810
1949	2,090
1950	3,620
1951	4,920
1952	4,090
1953	5,570
1954	9,400
1955	32,900
1956	8,710
1957	3,850
1958	4,970
1959	5,398
1960	4,780
1961	4,020
1962	5,790
1963	4,510
1964	5,520
1965	5,300

Problem 3. An electric scale gives a reading equal to the true weight plus a random error that is normally distributed with mean 0 and standard deviation $\sigma = .1$ mg. Suppose that the results of five successive weightings of the same object are as follows: 3.142, 3.163, 3.155, 3.150, 3.141.

- (a) Determine a 95 percent confidence interval estimate of the true weight.
- (b) Determine a 99 percent confidence interval estimate of the true weight.

Problem 4. The standard deviation of test scores on a certain achievement test is 11.3. If a random sample of 81 students had a sample mean score of 74.6, find a 90 percent confidence interval estimate for the average score of all students.

Problem 5. Suppose that a marketing research firm wants to conduct a survey to estimate the mean μ of the distribution of the amount spent on entertainment by each adult who visits a certain popular resort. The firm would like to estimate the mean of this distribution to within \$60 with 95% confidence. From data regarding past operations at the resort, it has been estimated that the standard deviation of the entertainment expenditures is no more than \$400. How large does the firm's sample size need to be?